SAVEETHA SCHOOL OF ENGINEERING

Assignment – 1

Course Code: ITA0603

Course Title: MACHINE LEARNING

Slot: C

Assignment Topic

Role of Machine Learning in Disease Prediction and Early Diagnosis

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Submitted By

Name: Pranav M V

Register Number: 192312358

Submitted To

Department of Computer Science and Engineering
Saveetha School of Engineering

1. Introduction

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables computers to learn patterns from data without being explicitly programmed. In healthcare, ML helps doctors predict diseases at an early stage by analyzing patient information such as age, blood pressure, glucose level, cholesterol, and other medical records.

Early disease prediction improves treatment success, reduces healthcare costs, and saves lives.

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2. Problem Statement

Many diseases such as diabetes, heart disease, and cancer are difficult to detect during their early stages because symptoms may not be visible.

Objective

- Predict whether a patient has diabetes using Machine Learning.
- Perform data preprocessing.
- Analyze the dataset using NumPy and Pandas.
- Visualize important features.
- Relate the solution to Unit I Machine Learning concepts.

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3. Machine Learning Concepts (Unit I)

Learning Problem

A learning problem consists of:

- Input: Patient medical records
- Output: Disease prediction (Positive/Negative)

Example:

Glucose	BMI	Age	Diabetes
120	28	40	No

Glucose	BMI	Age	Diabetes
180	35	50	Yes

Perspectives and Issues

Advantages

- Fast diagnosis
- High accuracy
- Supports doctors
- Detects hidden patterns

Issues

- Poor quality data
- Privacy concerns
- Bias in training data
- Overfitting

Concept Learning

Machine Learning learns concepts from examples.

Example

IF

Glucose > 140

AND

BMI > 30

THEN

Diabetes = Yes

Version Space and Candidate Elimination

Version Space contains all hypotheses consistent with training examples.

Candidate Elimination Algorithm removes inconsistent hypotheses until only the best hypotheses remain.

This helps reduce classification errors.

Inductive Bias

Machine Learning assumes unseen examples behave similarly to training data.

Example:

If many diabetic patients have

- High glucose
- High BMI

then future patients with similar values are likely diabetic.

Decision Tree Learning

Decision Trees classify patients using simple questions.

Example

```
           Glucose > 140?
            /        \
          No          Yes
         Healthy   BMI > 30?
                     /      \
                  No        Yes
             Low Risk   Diabetes
```

Decision Trees are popular in healthcare because they are easy to understand.

Representation Algorithm

Patient information is represented as numerical features.

Example

Feature	Value
Age	45
BMI	31
Glucose	180
Blood Pressure	80

Heuristic Space Search

Decision Trees use heuristic measures such as

- Information Gain
- Gini Index

to select the best feature for splitting data.

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4. Dataset

Dataset Used

Pima Indians Diabetes Dataset

Features

- Pregnancies
- Glucose
- Blood Pressure
- Skin Thickness
- Insulin
- BMI
- Diabetes Pedigree Function
- Age
- Outcome

Target

Outcome

0 = No Diabetes

1 = Diabetes

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5. Data Collection and Preprocessing

Libraries

```
import pandas as pd
import numpy as np
```

Load Dataset

```
df = pd.read_csv("diabetes.csv")
```

Display Data

```
print(df.head())
```

Check Missing Values

```
print(df.isnull().sum())
```

Basic Statistics

```
print(df.describe())
```

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6. Analysis using NumPy and Pandas

Average Glucose

```
print(df["Glucose"].mean())
```

Average BMI

```
print(df["BMI"].mean())
```

Diabetes Count

```
print(df["Outcome"].value_counts())
```

Correlation

```
print(df.corr())
```

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7. Visualization

Import Library

```
import matplotlib.pyplot as plt
```

Diabetes Distribution

```
df["Outcome"].value_counts().plot(kind="bar")
plt.title("Diabetes Distribution")
plt.xlabel("Outcome")
plt.ylabel("Count")
plt.show()
```

Glucose Histogram

```
plt.hist(df["Glucose"], bins=20)
plt.title("Glucose Distribution")
plt.xlabel("Glucose")
plt.ylabel("Frequency")
plt.show()
```

Correlation Heatmap

```
import seaborn as sns

sns.heatmap(df.corr(), annot=True)
plt.show()
```

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8. Expected Results

- High glucose strongly indicates diabetes.
- Higher BMI increases diabetes risk.
- Age also contributes to disease prediction.
- Decision Trees can classify patients effectively using these features.

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9. SDG Mapping

SDG 3 – Good Health and Well-being

Machine Learning contributes by:

- Early disease detection
- Reducing mortality
- Improving healthcare quality
- Supporting doctors in clinical decision-making
- Enabling preventive healthcare

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10. Advantages

- Faster diagnosis
- Higher prediction accuracy
- Cost-effective
- Supports personalized medicine
- Reduces manual errors

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11. Limitations

- Requires quality medical data
- Risk of biased predictions

- Privacy and security concerns
- Models require regular updates
- Cannot replace medical professionals

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12. Conclusion

Machine Learning plays a significant role in disease prediction and early diagnosis by analyzing medical data to identify patterns associated with diseases such as diabetes. Techniques like **Concept Learning**, **Decision Tree Learning**, **Version Spaces**, **Candidate Elimination**, and **Heuristic Search** enable accurate and efficient prediction. With proper preprocessing, analysis using **NumPy** and **Pandas**, and effective visualization, ML models can support healthcare professionals in making informed decisions. This application aligns with **Sustainable Development Goal (SDG) 3: Good Health and Well-being**, promoting improved healthcare outcomes and early intervention.